Akoizo Original Practice - Form 3

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A body orbits the Sun at 9 AU. What is its period?

A. 27 years

B. 9 years

C. 81 years

D. 3 years

Answer: _____

2. (2 points) Which mission returned a sample of asteroid Bennu?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. OSIRIS-REx

B. Rosetta

C. Dragonfly

D. Kepler

Answer: _____

3. (2 points) If a star's surface temperature doubles, its emitted power per area changes by:

A. 4 times

B. 8 times

C. 16 times

D. 2 times

Answer: _____

4. (2 points) Silica nanoparticles in Enceladus's plume suggest:

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Methane lakes

B. Recent asteroid impacts

C. Hydrothermal activity at the seafloor

D. A thick atmosphere

Answer: _____

5. (2 points) Hydrogen in a plume along with carbon dioxide is evidence for:

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Chemical disequilibrium as an energy source

B. A thick atmosphere

C. A magnetic field

D. Surface lakes

Answer: _____

6. (2 points) Which mission measures gas escaping from Mars's upper atmosphere?

A. MAVEN

B. Perseverance

C. DAVINCI

D. Venus Express

Answer: _____

7. (2 points) Why are habitable-zone planets around red dwarfs likely tidally locked?

A. They orbit very close to their dim stars

B. They are very far from their stars

C. Red dwarfs have no gravity

D. They have thick rings

Answer: _____

8. (4 points) Explain Kepler's second law in terms of orbital speed.

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

Answer: _____
_____
_____

9. (4 points) Why are samples returned to Earth more valuable than remote measurements?

Answer: _____
_____
_____

10. (4 points) Explain how albedo affects a planet's equilibrium temperature.

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

Answer: _____
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_____

11. (4 points) Explain why Enceladus's plume is a convenient way to study its ocean.

Answer: _____
_____
_____

12. (4 points) How do salts help liquid water exist in cold places?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun’s habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

Answer: _____
_____
_____

13. (4 points) Compare the goals of DAVINCI and VERITAS.

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

Answer: _____
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14. (4 points) What does the TRAPPIST-1 system show about habitability around red dwarfs?

Answer: _____
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15. (4 points) Describe what tidal locking means for climate on an exoplanet.

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth’s mass and 2 times Earth’s radius. Compare its surface gravity and escape velocity with Earth’s.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

Answer: _____
_____
_____

16. (4 points) Describe one piece of evidence that Bennu's parent body had liquid water.

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

Answer: _____
_____
_____

17. (6 points) Solve this case and explain your reasoning: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.

Answer: _____
_____
_____

18. (6 points) Solve this case and explain your reasoning: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?

Answer: _____
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